Fusion Information Bottleneck-Driven Vision Transformer Model for Diagnosis of Retinal Diseases via Internet of Medical Things

Jian Hu, Rui Ma, Ning Yu, Shuo Wang, Bingshuo Li, Hanyi Ren, Lan Zhang, Qian Li, Xinji Yang, Yongcai Wang, and Yueyue Li

***Abstract*—Ocular diseases are among the leading causes of visual impairment and blindness worldwide, and early, accurate diagnosis is crucial for preventing disease progression. In recent years, deep learning methods based on fundus images have achieved remarkable progress in intelligent diagnosis, yet they still face challenges such as severe overfitting, redundant features, and insufficient capability in fine-grained lesion recognition. These issues become more pronounced in the Internet of Medical Things (IoMT) scenario, further limiting the generalization and practicality of the models. To address this, this article proposes a vision transformer model incorporating information bottleneck and contrastive learning mechanisms (FIBCL-ViT) for multiclass fundus disease diagnosis. Using a vision transformer (ViT) as the backbone, the model introduces a variational information bottleneck (VIB) module to compress redundant features and highlight task-relevant information while integrating a contrastive learning strategy to enhance feature discriminability, thereby improving the model's generalization ability and robustness. During training, the model jointly optimizes cross-entropy loss, contrastive loss, and the information bottleneck (IB) regularization term to achieve efficient learning. The experimental results demonstrate that the proposed FIBCL-ViT model outperforms mainstream comparison methods across multiple public fundus image datasets, achieving superior performance in terms of accuracy, area under the receiver operating characteristic (AUC), and other metrics. For the 7:3 training–testing split, the proposed method achieves an accuracy of 0.9376, a precision of 0.9384, a recall of 0.9374, and an AUC of 0.9902 on public fundus image datasets. This study provides a novel solution for automated screening and remote intelligent diagnosis of ophthalmic diseases, showing strong potential for clinical applications.**

***Index Terms*—Contrastive learning, fundus disease diagnosis, information bottleneck (IB), Internet of Medical Things (IoMT), vision transformer (ViT).**

Received 20 December 2025; revised 11 February 2026; accepted 24 February 2026. Date of publication 6 March 2026; date of current version 11 May 2026. This work was supported in part by the National Key Research and Development Program of China under Grant 2024YFB4710205 and in part by the National Natural Science Foundation of China under Grant 61972404. *(Jian Hu, Rui Ma, and Ning Yu contributed equally to this work.) (Corresponding authors: Yueyue Li; Xinji Yang; Yongcai Wang.)*

Jian Hu, Rui Ma, Xinji Yang, and Yueyue Li are with the Senior Department of Ophthalmology, Chinese PLA General Hospital, Beijing 100039, China (e-mail: hujian301@126.com; 56771651@qq.com; yangxinji68@sina.com; 13466365403@163.com).

Ning Yu is with the Senior Department of Otolaryngology-Head and Neck Surgery, Chinese PLA General Hospital, Beijing 100048, China (e-mail: yuning@301hospital.org).

Shuo Wang and Yongcai Wang are with the School of Information, Renmin University of China, Beijing 100872, China (e-mail: shuowang18@ruc.edu.cn; ycw@ruc.edu.cn).

Bingshuo Li is with the School of Optical and Electronic Information, Huazhong University of Science and Technology, Wuhan, Hubei 430074, China (e-mail: U202314782@hust.edu.cn).

Hanyi Ren is with the School of North Alabama International College of Engineering and Technology, Guizhou University, Guyang, Guizhou 550025, China (e-mail: naie.hyren23@gzu.edu.cn).

Lan Zhang is with the Senior Department of General Surgery, Chinese PLA General Hospital, Beijing 100853, China (e-mail: zhanglan301@163.com).

Qian Li is with the Computing and Science Laboratory, Mitomed Technology Company Ltd., Beijing 100000, China (e-mail: qianli301@126.com).

Digital Object Identifier 10.1109/JIOT.2026.3671288

I. Introduction

OCULAR diseases, such as diabetic retinopathy [1], [2], glaucoma [3], [4], [5], and age-related macular degeneration [6], [7], are among the leading causes of visual impairment and blindness worldwide. Their incidence continues to rise annually, imposing a heavy burden on public health and socioeconomic systems. Early and accurate diagnosis plays a decisive role in preventing disease progression and improving treatment outcomes. However, traditional diagnostic approaches [8] rely heavily on the expertise and manual interpretation of ophthalmologists, which are time-consuming, labor-intensive, and often insufficient to meet the demands of large-scale screening and timely diagnosis, particularly in resource-limited regions.

With the rapid advancement of deep learning [9], [10], and computer vision [11], [12], automated ocular disease recognition has emerged as a key research direction in medical image analysis. In particular, intelligent diagnosis based on fundus images [13], [14], [15] has attracted significant attention due to its noninvasive, low-cost, and information-rich nature. This approach is regarded as an effective pathway to enable large-scale ophthalmic screening and support clinical decision-making. By integrating deep learning models into the diagnostic process, physicians can benefit from efficient and precise auxiliary tools that reduce workload, improve diagnostic consistency, and provide patients with critical access to early treatment opportunities.

Furthermore, with the fast development of the Internet of Medical Things (IoMT) [16], fundus imaging devices can be seamlessly connected with intelligent diagnostic systems, enabling patients to undergo eye examinations and automated analysis in primary care facilities or even remote settings. This integration of deep learning and IoMT holds great potential to overcome the uneven distribution of medical resources, enhance the accessibility and timeliness of ophthalmic screening, and ultimately deliver greater value in both public health management and clinical practice.

Nevertheless, current research and applications still face significant challenges. Fundus images contain abundant details, such as vascular structures, background textures, and imaging noise, yet not all of these details are directly relevant to disease diagnosis. Conventional models often "memorize" such irrelevant features, leading to good performance on training datasets but performance degradation in real clinical scenarios or cross-domain datasets, i.e., the problems of overfitting and insufficient generalization. In addition, many ocular diseases exhibit extremely subtle differences in fundus images. For instance, the lesion characteristics between different stages of diabetic retinopathy, or the slight changes in the early stages of macular degeneration, are difficult for traditional models to capture effectively. Existing approaches often lack the ability to model fine-grained differences in the feature embedding space, which limits their robustness and discriminative power in complex clinical environments.

In this study, we propose a contrastive vision transformer model with information bottleneck, designed to address the challenges of overfitting, redundant features, and class imbalance in fundus disease diagnosis under IoMT scenarios. The core idea of vision transformer model incorporating information bottleneck and contrastive learning mechanisms (FIBCL-ViT) is to integrate the global modeling capability of the vision transformer (ViT) [17], [18], the feature compression mechanism of the information bottleneck (IB) [19], [20], and the discriminative enhancement strategy of contrastive learning [21], [22], thereby improving the model's generalization and robustness while maintaining high accuracy.

In particular, the FIBCL-ViT model first applies diverse data augmentation operations to the input fundus images, including random cropping, rotation, and color perturbation to simulate variations in real-world acquisition conditions and generate two sets of independent augmented samples. These augmented samples are then input into a ViT-based feature extraction network. By leveraging self-attention, the ViT encoder models global dependencies and simultaneously captures fine-grained lesion features and overall structural information, providing a rich representational foundation for subsequent tasks.

After obtaining the initial feature representations, the model incorporates an IB module. Through the variational IB (VIB) approach, the latent representations are forced to be compressed such that only label-relevant discriminative information is retained, while task-irrelevant details (e.g., background textures and noise) are discarded. With the IB constraint, the model learns to suppress reliance on background or irrelevant details during training, thereby reducing overfitting and enhancing cross-domain generalization.

On top of the compressed representations, FIBCL-ViT further introduces a contrastive learning mechanism. By optimizing a contrastive loss, the model encourages representations of the same class to cluster closely in the embedding space, while pushing apart those from different classes. This not only strengthens the model's ability to capture subtle lesion differences but also significantly improves discriminative performance under class imbalance. In particular, for minority-class recognition, contrastive learning enhances feature separability and reduces bias toward high-frequency classes.

Finally, the compressed and contrastively optimized features are input into a classifier to predict specific ocular disease categories. During training, a joint optimization strategy is adopted, combining supervised cross-entropy loss for classification, contrastive loss for enhanced discriminability, and the IB regularization term. Together, these components enable FIBCL-ViT to reduce feature redundancy, preserve interclass discriminability, and improve stability and generalization in clinical applications.

The main contributions of this article are as follows.

1) For IoMT scenarios, we propose the FIBCL-ViT model for fundus disease diagnosis.
2) By imposing an IB constraint in the feature embedding space, the FIBCL-ViT model can automatically filter redundant details and highlight task-relevant features. Meanwhile, the integration of a contrastive loss function enhances interclass separability, making the model more robust in disease classification.
3) We validate that the FIBCL-ViT model outperforms mainstream methods in terms of classification accuracy, generalization ability, and cross-domain robustness, demonstrating its potential application value in clinical decision support and real-world deployment.

The remainder of this article is organized as follows. Section II introduces related work. Section III presents the implementation details of the FIBCL-ViT algorithm. Section IV reports the experimental results under various settings. Section V concludes this article.

II. Related Work

In recent years, the application of deep learning in ophthalmic image analysis has attracted widespread attention and achieved remarkable progress in automated disease detection and classification. Researchers have proposed various approaches for different types of ophthalmic images—such as OCT, fundus photography, and AS-OCT—as well as for different diseases, including diabetic retinopathy, glaucoma, macular degeneration, and corneal disorders, providing valuable References for clinical decision support.

For example, Umer et al. [23] addressed the limitations of ophthalmologists' manual interpretation of OCT images, which is prone to subjectivity and error, by proposing an automated ocular disease detection method based on deep feature fusion and selection. Their approach combined features extracted by an improved AlexNet and ResNet-50, which were fused serially and refined through feature selection before being input into multiple machine learning classifiers, enabling the recognition of four categories of ocular diseases.

In the context of diabetes-related eye disease detection, Albelaihi and Ibrahim [24] highlighted the fact that diabetic retinopathy, diabetic macular edema, glaucoma, and cataract are becoming major global causes of vision loss, while existing methods still suffer from insufficient accuracy. To address this, they proposed a multiclass deep learning framework, DeepDiabetic. Using six publicly available fundus image datasets (a total of 1228 images), they systematically evaluated five architectures, including EfficientNet, VGG16, and ResNet.

Similarly, focusing on OCT images, Adel et al. [25] tackled the limited accuracy of traditional deep learning methods in OCT image classification by introducing a model that combines transfer learning with a classification hinge loss function for the automatic detection of CNV, drusen, and normal retina. The experimental results demonstrated outstanding performance on a large-scale OCT dataset, where the Xception model achieved an accuracy of 98%, significantly outperforming existing methods.

In addition, Glaret subin and Muthukannan [26] focused on the early detection of age-related eye diseases and proposed a CNN-based multidisease detection method. Their approach employed maximum entropy transformation for image preprocessing, enhanced CNN feature extraction and hyperparameter tuning through optimization algorithms, and finally integrated a multiclass SVM classifier to automatically recognize AMD, cataract, diabetic retinopathy, and glaucoma, achieving high diagnostic performance on the ODIR dataset.

In resource-limited developing countries, Bernabé et al. [27] addressed the difficulty of timely ophthalmic diagnosis by introducing a CNN-based intelligent pattern classification algorithm for the automatic detection of glaucoma and diabetic retinopathy from fundus images. The experimental results showed near-perfect performance across multiple evaluation metrics, with an accuracy of 99.89%, confirming its reliability and feasibility for automated eye disease detection.

Furthermore, Ting et al. [28] extended their research to large-scale population screening by proposing and evaluating a deep learning system capable of automatically detecting referable diabetic retinopathy, vision-threatening diabetic retinopathy, possible glaucoma, and AMD. Trained and validated on more than 490 000 retinal images, the system demonstrated excellent diagnostic performance in the Singapore National Diabetic Retinopathy Screening Program and a multiethnic diabetic cohort, laying a foundation for large-scale clinical deployment.

Beyond retinal diseases, Elsawy et al. [29] targeted common corneal conditions such as dry eye syndrome and keratoconus, and proposed a multidisease deep learning diagnostic network based on AS-OCT images. Trained on 158 220 images, the model achieved superior performance on an independent test set, highlighting the potential of deep learning for automated diagnosis of corneal diseases.

Despite these advances, existing methods still face significant challenges. Fundus images often contain substantial task-irrelevant information, such as background textures and imaging noise, which can lead to overfitting and reduced robustness. Moreover, interdisease differences are often subtle, and both CNN-based and standard transformer-based models may struggle to learn compact and discriminative representations without additional regularization.

Motivated by these limitations, this study proposes the FIBCL-ViT framework, which builds upon transformer-based architectures while explicitly incorporating the IB principle and contrastive learning. Compared with existing CNN-based and transformer-based approaches, the proposed method aims to reduce feature redundancy, enhance discriminative representation learning, and improve robustness and generalization in retinal disease diagnosis, particularly in IoMT-oriented application scenarios.

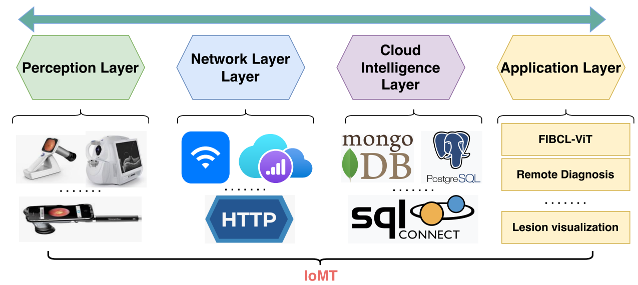

Fig. 1. IoMT-based architecture for deploying the proposed FIBCL-ViT model.

III. Method

In this study, we propose the FIBCL-ViT, designed to address issues such as overfitting, feature redundancy, and class imbalance in the diagnosis of fundus diseases within IoMT scenarios.

As shown in Fig. 1, the FIBCL-ViT model is deployed within an IoMT architecture, which consists of four functional modules: the perception layer, network layer, cloud server layer, and application layer. This architecture is intended to enable the automatic acquisition, transmission, storage, and intelligent diagnosis of fundus images.

The perception layer comprises various fundus image acquisition devices, including portable fundus cameras, OCT scanners, and smartphone-compatible ophthalmoscopes, to achieve real-time collection of users' fundus images. The network layer relies on wireless communication protocols such as Wi-Fi and 4G/5G to securely and efficiently transmit the collected image data to a remote cloud platform. The cloud server layer consists of a structured image storage system and database management modules, supporting large-scale storage, indexing, and retrieval of fundus images.

Finally, the application layer integrates the proposed deep learning diagnostic model FIBCL-ViT along with modules for remote-assisted diagnosis, enabling automatic analysis and classification of uploaded images while providing intelligent diagnostic suggestions and visualization support for clinicians. This IoMT architecture not only realizes a full-process closed loop for fundus image diagnosis but also provides the infrastructure necessary for efficient and scalable intelligent eye disease screening. In this workflow, the FIBCL-ViT model serves as the core decision-making engine at the application layer, directly operating on the transmitted fundus images and generating diagnostic outputs that can be accessed by clinicians or healthcare systems for subsequent clinical interpretation and decision support.

The design philosophy of the FIBCL-ViT model is to combine the global modeling capability of ViT, the feature

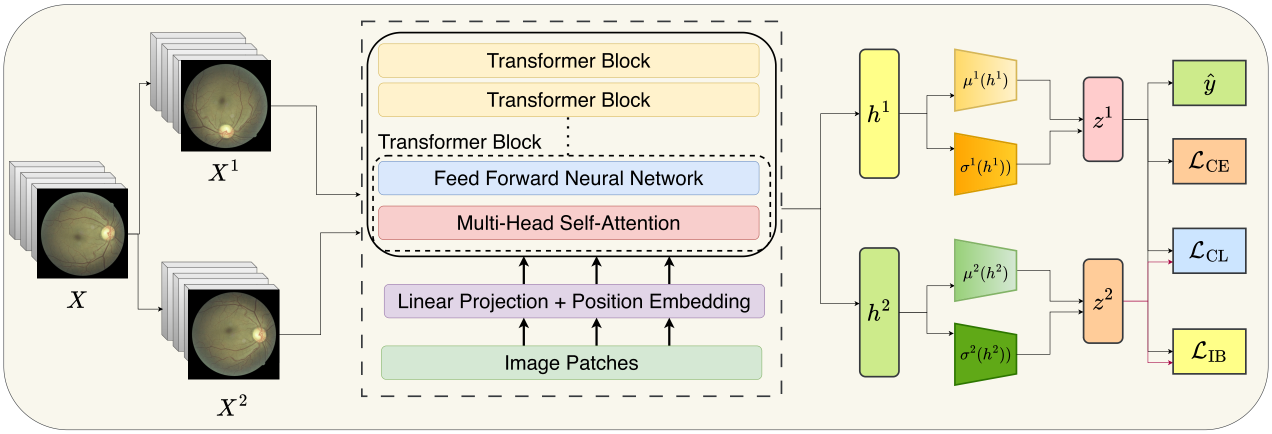

Fig. 2. Implementation details of the proposed FIBCL-ViT model.

compression mechanism of the IB, and the discriminative enhancement strategy of contrastive learning, thereby significantly improving the model's generalization ability and robustness while maintaining high accuracy.

In particular, the implementation details of the FIBCL-ViT model are as shown in Fig. 2.

As shown in (1) and (2), the FIBCL-ViT model first performs data augmentation on the fundus image X, generating two independently augmented samples X^1 and X^2

$$X^1 = \mathcal{T}_1(X) \tag{1}$$
$$X^2 = \mathcal{T}_2(X) \tag{2}$$

where $\mathcal{T}_1$ and $\mathcal{T}_2$ denote two different data augmentation strategies.

Next, as in (3) and (4), the FIBCL-ViT model feeds X^1 and X^2 into a ViT-based feature extraction network, obtaining representations h^1 and h^2, respectively,

$$h^1 = f_{\mathrm{ViT}}\left(X^1;\theta\right) \tag{3}$$
$$h^2 = f_{\mathrm{ViT}}\left(X^2;\theta\right). \tag{4}$$

These representations capture both fine-grained lesion features and the overall structural information of the image, where $f_{\mathrm{ViT}}(\cdot;\theta)$ denotes a ViT encoder with shared parameters θ.

Subsequently, after obtaining h^1 and h^2, the FIBCL-ViT model introduces the **IB module**. As in (5) and (6), based on h^1 and h^2, the model uses conditional Gaussian distributions to model the compressed representations z^1 and z^2

$$q(z^1|h^1) = \mathcal{N}\left(z^1;\mu^1\left(h^1\right),\sigma^1\left(h^1\right)\right) \tag{5}$$
$$q(z^2|h^2) = \mathcal{N}\left(z^2;\mu^2\left(h^2\right),\sigma^2\left(h^2\right)\right). \tag{6}$$

Moreover, as in (7) and (8), the model samples from these distributions using the reparameterization trick to obtain z^1 and z^2

$$z^1 = \mu^1\left(h^1\right) + \sigma^1\left(h^1\right) \odot \epsilon^1 \tag{7}$$
$$z^2 = \mu^2\left(h^2\right) + \sigma^2\left(h^2\right) \odot \epsilon^2 \tag{8}$$

where $\epsilon^i \sim \mathcal{N}(0, I)$ and $\odot$ denotes elementwise multiplication.

These compressed representations retain only the discriminative information relevant to disease labels while discarding task-irrelevant redundant details, such as background textures and noise.

Next, via the IB loss in (9), the model minimizes the redundant mutual information between the latent variables and the input images

$$\mathcal{L}_{\mathrm{IB}} = \beta \sum_{i=1,2} \mathrm{KL}\left(q\left(z^i \middle| h^i\right) \middle\| p\left(z^i\right)\right) \tag{9}$$

where $\mathrm{KL}(\cdot\|\cdot)$ denotes the KL divergence, $p(z^i) = \mathcal{N}(0, I)$ is the prior distribution, and β is a balancing parameter.

Under the IB constraint, the model can automatically suppress the tendency to overly rely on background or irrelevant details during training, thereby reducing overfitting and enhancing cross-domain generalization.

Then, as in (10), on top of the compressed latent representations, FIBCL-ViT further introduces a contrastive learning mechanism

$$\mathcal{L}_{\mathrm{CL}} = -\log \frac{\exp\left(\mathrm{sim}\left(z^1, z^2\right)/\tau\right)}{\sum_{k=1}^{N} \exp\left(\mathrm{sim}\left(z^1, z^k\right)/\tau\right)} \tag{10}$$

where $\mathrm{sim}(\cdot,\cdot)$ denotes the cosine similarity function, τ is the temperature parameter, and N is the batch size.

Through the contrastive loss, the model pulls representations of the same class closer in the embedding space while pushing representations of different classes apart. This not only strengthens the model's ability to capture subtle lesion differences but also significantly improves discriminative performance under class imbalance. Particularly for minority classes, contrastive learning effectively enhances feature separability, reducing the model's bias toward high-frequency classes.

Next, as in (11) and (12), the model feeds the features, optimized through IB compression and contrastive learning, into the classifier to predict specific fundus disease categories

$$\bar{z} = \frac{z^1 + z^2}{2} \tag{11}$$
$$\hat{y} = \mathrm{softmax}(W\bar{z} + b) \tag{12}$$

where $W \in \mathbb{R}^{C\times d'}$ and $b \in \mathbb{R}^{C}$ are classifier parameters, and C is the number of disease categories.

The following equation shows the cross-entropy loss used for classification:

$$\mathcal{L}_{\text{CE}} = -\sum_{c=1}^{C} y_c \log(\hat{y}_c) \tag{13}$$

where y is the one-hot encoding of the ground-truth label.

During training, the model adopts the joint optimization strategy in the following equation:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda\mathcal{L}_{\text{CL}} + \beta\mathcal{L}_{\text{IB}}. \tag{14}$$

This strategy combines the supervised classification cross-entropy loss, the discriminative contrastive loss, and the IB regularization. Together, these three components allow FIBCL-ViT to reduce feature redundancy while maintaining interclass discriminability, improving stability and generalization in clinical applications.

Here, λ and β are weighting coefficients for the contrastive loss and IB loss, balancing the importance of the three optimization objectives.

Algorithm 1 FIBCL-ViT Training Procedure

Require: Fundus image dataset $\mathcal{D} = \{(X_i, y_i)\}_{i=1}^{N}$, batch size B, learning rate η, IB weight β, contrastive weight λ, number of epochs T

Initialize ViT encoder $f_{\text{ViT}}(\cdot;\theta)$, classifier parameters W, b

for epoch = 1 to T **do**

  for each mini-batch $\{X_i, y_i\}_{i=1}^{B}$ from $\mathcal{D}$ **do**

    Apply two random augmentations: $X_i^1 = \mathcal{T}_1(X_i)$, $X_i^2 = \mathcal{T}_2(X_i)$

    Extract ViT representations: $h_i^1 = f_{\text{ViT}}(X_i^1;\theta)$, $h_i^2 = f_{\text{ViT}}(X_i^2;\theta)$

    Compute IB compressed features using reparameterization:

    $z_i^1 = \mu^1(h_i^1) + \sigma^1(h_i^1) \odot \epsilon_i^1$, $\epsilon_i^1 \sim \mathcal{N}(0, I)$

    $z_i^2 = \mu^2(h_i^2) + \sigma^2(h_i^2) \odot \epsilon_i^2$, $\epsilon_i^2 \sim \mathcal{N}(0, I)$

    Compute joint feature: $\bar{z}_i = (z_i^1 + z_i^2)/2$

    Predict disease probabilities: $\hat{y}_i = \text{softmax}(W\bar{z}_i + b)$

    Compute losses:

    $\mathcal{L}_{\text{CE}} = -\sum_c y_{i,c} \log(\hat{y}_{i,c})$

    $\mathcal{L}_{\text{IB}} = \sum_{j=1,2} \text{KL}(q(z_i^j|h_i^j) \| p(z_i^j))$

    $\mathcal{L}_{\text{CL}} = -\log \frac{\exp(\text{sim}(z_i^1, z_i^2)/\tau)}{\sum_{k=1}^{B} \exp(\text{sim}(z_i^1, z_k^2)/\tau)}$

    Compute total loss: $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda\mathcal{L}_{\text{CL}} + \beta\mathcal{L}_{\text{IB}}$

    Update parameters θ, W, b using gradient descent: $\theta \leftarrow \theta - \eta\nabla_\theta\mathcal{L}_{\text{total}}$, etc.

  end for

end for

Ensure: Trained FIBCL-ViT model

Algorithm 1 presents the pseudocode of FIBCL-ViT. By skillfully integrating the global modeling advantage of ViT, the feature compression capability of the IB, and the discriminative enhancement of contrastive learning, the FIBCL-ViT model achieves both high accuracy and strong generalization in fundus disease diagnosis, providing an efficient and robust deep learning solution for clinical intelligent diagnosis.

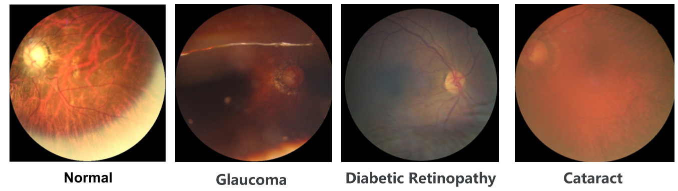

Fig. 3. Representative samples of fundus images for each disease category in the dataset.

IV. Experiment and Discussion

A. Dataset

The fundus image dataset used in this study includes four disease categories: cataract (1038 images), diabetic retinopathy (1098 images), glaucoma (1007 images), and normal cases (1074 images), ensuring an overall balanced distribution. The images were collected from multiple public databases, including IDRiD, Ocular Disease Recognition, and HRF, thereby ensuring diversity in disease types, image quality, and acquisition conditions. To standardize the input format, all images were preprocessed before being input into the model, including resizing to 224×224 and pixel value normalization, in order to meet the input requirements of the backbone network. Fig. 3 shows example samples of each eye disease category in the dataset.

For dataset partitioning, a stratified sampling strategy was employed to divide the data into training and testing sets using three different ratios: 3:7, 5:5, and 7:3. This approach ensured a uniform distribution of all categories across the subsets under each split setting. The training sets were used for model parameter learning, while the corresponding testing sets were reserved for performance evaluation and generalization assessment.

B. Experimental Setup

The experiments in this study were conducted on a high-performance computing platform equipped with an NVIDIA GeForce RTX 3090Ti GPU (16-GB memory) and an Intel Core i9 processor, running on Ubuntu. The model was implemented using the PyTorch deep learning framework, primarily in Python, with image preprocessing and result visualization assisted by commonly used libraries such as OpenCV and Matplotlib.

To comprehensively evaluate the performance of the proposed model, several widely used classification metrics were employed, including accuracy, precision, recall, and the area under the receiver operating characteristic curve (AUC).

For benchmarking, multiple state-of-the-art image classification models were selected as baselines for comparison, including DenseNet, ViT, Swin Transformer V1, Swin Transformer V2, and ConvNeXt V2.

During training, the FIBCL-ViT model was optimized using AdamW with an initial learning rate of 1×10^{-4}, a batch size of 16, and ten training epochs.

Table I summarizes the hardware and software environment, evaluation metrics, baseline models, and the key hyperparameter settings of the proposed FIBCL-ViT model.

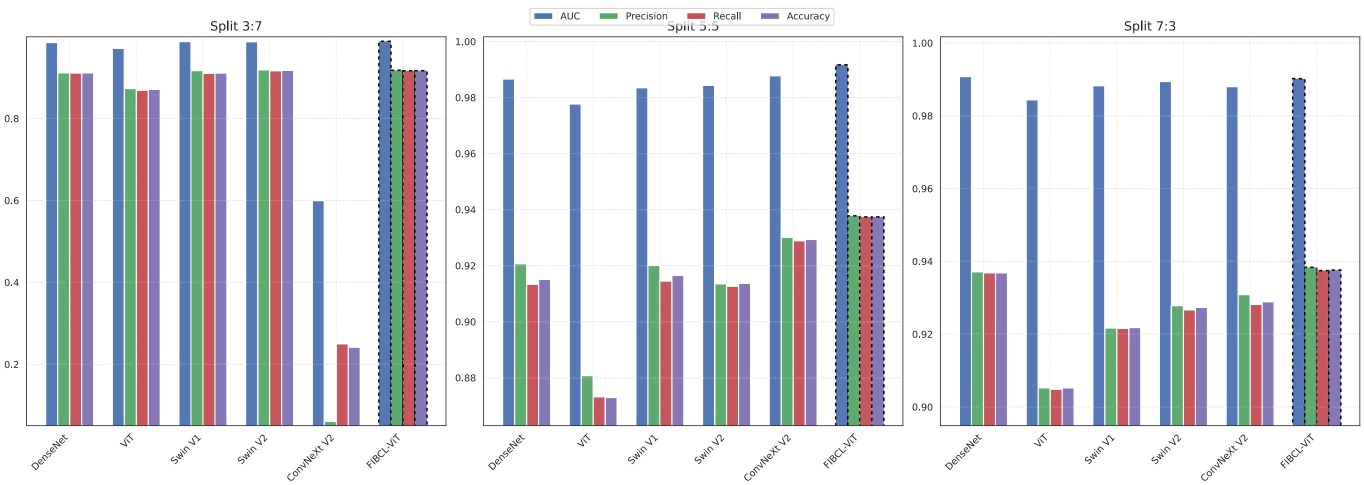

Fig. 4. Performance comparison of different models across three training–testing splits (3:7, 5:5, and 7:3) on four evaluation metrics (AUC, precision, recall, and accuracy).

TABLE I
HARDWARE AND SOFTWARE ENVIRONMENT, EVALUATION METRICS, BASELINE MODELS, AND KEY HYPERPARAMETER SETTINGS

Category	Specification
Hardware Environment	GPU: NVIDIA GeForce RTX 3090Ti CPU: Intel Core i9 processor Operating System: Ubuntu Memory: 16 GB VRAM
Software Environment	Framework: PyTorch Programming Language: Python Image Processing: OpenCV Visualization: Matplotlib
Evaluation Metrics	Accuracy Precision Recall AUC
Baseline Models	DenseNet ViT Swin Transformer V1 Swin Transformer V2 ConvNeXt V2
FIBCL-ViT Hyperparameters	Optimizer: AdamW Initial Learning Rate: 1×10^{-4} Batch Size: 16 Training Epochs: 10

TABLE II
PERFORMANCE COMPARISON UNDER 3:7 TRAINING–TESTING SPLIT

Model	AUC	Precision	Recall	Accuracy
DenseNet	0.9853	0.9116	0.9109	0.9112
ViT	0.9712	0.8731	0.8691	0.8709
Swin V1	0.9878	0.9170	0.9105	0.9109
Swin V2	0.9875	0.9188	0.9166	0.9173
ConvNeXt V2	0.5996	0.0605	0.2500	0.2419
FIBCL-ViT (Ours)	**0.9883**	**0.9175**	**0.9164**	**0.9167**

C. Overall Performance Evaluation

To comprehensively evaluate the performance of the proposed FIBCL-ViT model, we systematically compared it with several representative mainstream image classification architectures under three different training-to-testing split ratios: 3:7, 5:5, and 7:3. The compared models include DenseNet, ViT, Swin Transformer V1, Swin Transformer V2, and ConvNeXt V2. These three split settings simulate practical scenarios with limited training data, moderate data availability, and sufficient training samples, respectively, allowing a thorough assessment of model robustness and generalization ability. The quantitative results under the three settings are reported in Tables II–IV and Fig. 4.

Under the data-scarce scenario with a training-to-testing split of 3:7, the proposed FIBCL-ViT model achieved the best overall performance among all compared methods, with an AUC of 0.9883, a precision of 0.9175, a recall of 0.9164, and an accuracy of 0.9167.

In comparison, DenseNet obtained an AUC of 0.9853, a precision 0.9116, a recall 0.9109, and an accuracy 0.9112, while ViT achieved 0.9712, 0.8731, 0.8691, and 0.8709, respectively. Swin Transformer V1 and Swin Transformer V2 showed competitive results, with AUC values of 0.9878 and 0.9875, and accuracy values of 0.9109 and 0.9173, respectively. Notably, ConvNeXt V2 exhibited a clear performance degradation under this setting, with an AUC of only 0.5996 and an accuracy of 0.2419, indicating limited robustness in extremely data-limited conditions.

Overall, FIBCL-ViT consistently outperformed the best competing model (Swin V2) in terms of AUC and recall, demonstrating stronger stability and discriminative ability when training data are scarce.

When the training-to-testing ratio was set to 5:5, representing a balanced data regime, the performance of all models improved. Under this setting, FIBCL-ViT again achieved the highest scores across all evaluation metrics, reaching an AUC of 0.9917, a precision of 0.9378, a recall of 0.9374, and an accuracy of 0.9374.

Among the baseline models, DenseNet achieved an accuracy of 0.9151, while ViT remained relatively weaker with an accuracy of 0.8729. Swin Transformer V1 and Swin Transformer V2 achieved comparable results, with accuracy values of 0.9165 and 0.9137, respectively. ConvNeXt V2 showed a substantial performance recovery compared to the 3:7 split, achieving an AUC of 0.9878 and an accuracy of 0.9294, but still lagged behind FIBCL-ViT by 0.0080 in accuracy.

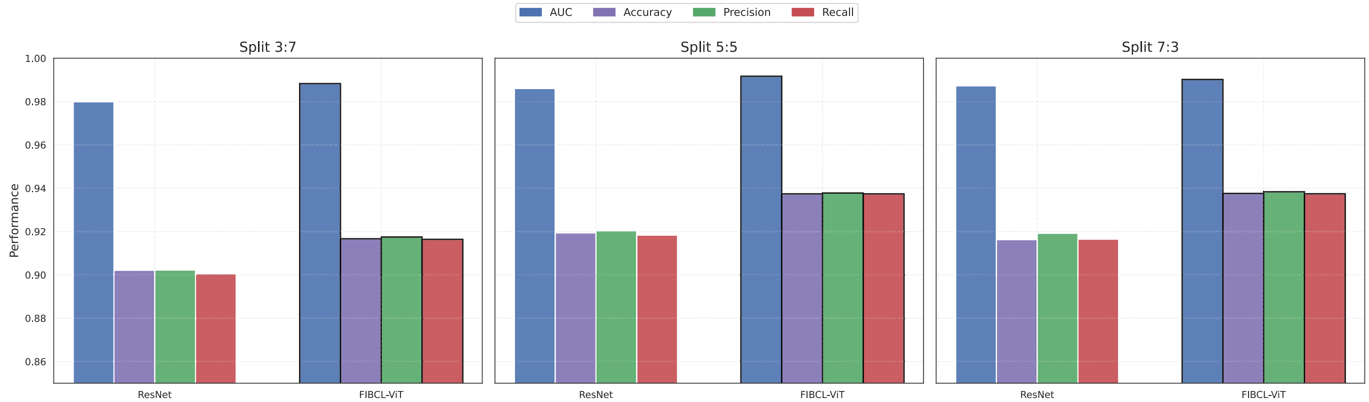

Fig. 5. Performance comparison between ViT-based and CNN-based as feature extraction modules.

TABLE III

PERFORMANCE COMPARISON UNDER 5:5 TRAINING–TESTING SPLIT

Model	AUC	Precision	Recall	Accuracy
DenseNet	0.9866	0.9207	0.9134	0.9151
ViT	0.9777	0.8808	0.8732	0.8729
Swin V1	0.9835	0.9201	0.9145	0.9165
Swin V2	0.9843	0.9135	0.9127	0.9137
ConvNeXt V2	0.9878	0.9301	0.9289	0.9294
FIBCL-ViT (Ours)	**0.9917**	**0.9378**	**0.9374**	**0.9374**

TABLE IV

PERFORMANCE COMPARISON UNDER 7:3 TRAINING–TESTING SPLIT

Model	AUC	Precision	Recall	Accuracy
DenseNet	0.9908	0.9371	0.9368	0.9368
ViT	0.9844	0.9053	0.9049	0.9052
Swin V1	0.9883	0.9217	0.9216	0.9218
Swin V2	0.9894	0.9278	0.9266	0.9273
ConvNeXt V2	0.9880	0.9309	0.9282	0.9289
FIBCL-ViT (Ours)	**0.9902**	**0.9384**	**0.9374**	**0.9376**

These results indicate that FIBCL-ViT maintains superior performance and stability even when competing models benefit from increased training data.

Under the 7:3 split, where a larger proportion of data is available for training, the overall performance of all models further improved. In this data-abundant scenario, FIBCL-ViT continued to demonstrate the best performance, achieving an AUC of 0.9902, a precision of 0.9384, a recall of 0.9374, and an accuracy of 0.9376.

DenseNet achieved an accuracy of 0.9368, closely followed by ConvNeXt V2 with 0.9289. Swin Transformer V2 and Swin Transformer V1 reached accuracy values of 0.9273 and 0.9218, respectively, while ViT remained the weakest performer with an accuracy of 0.9052.

Although the performance gap between models narrowed as training data increased, FIBCL-ViT consistently preserved a measurable advantage in AUC and recall, suggesting stronger capability in capturing fine-grained discriminative features and maintaining robust generalization under sufficient data conditions.

TABLE V

ABLATION STUDY: VIT BACKBONE VERSUS RESNET BACKBONE IN THE FIBCL FRAMEWORK

Backbone	Split	AUC	Precision	Recall	Accuracy
ResNet	3:7	0.9799	0.9023	0.9005	0.9021
	5:5	0.9860	0.9204	0.9184	0.9194
	7:3	0.9873	0.9192	0.9165	0.9163
FIBCL-ViT	3:7	**0.9883**	**0.9175**	**0.9164**	**0.9167**
	5:5	**0.9917**	**0.9378**	**0.9374**	**0.9374**
	7:3	**0.9902**	**0.9384**	**0.9374**	**0.9376**

In summary, the FIBCL-ViT model outperformed existing mainstream models across all three split settings and maintained stable performance even with limited training data, validating its excellent robustness and generalization capability. This performance improvement is primarily attributed to the IB constraint mechanism introduced in the model design, which effectively filters redundant features and highlights key information relevant to disease classification, as well as the contrastive learning strategy, which further enhances the discrimination between interclass features, enabling the model to perform more excellently in multiclass diagnostic tasks.

D. Ablation Study

In this section, we conducted ablation experiments to evaluate the contribution of the feature extraction backbone in the proposed FIBCL-ViT framework.

In particular, we replaced the original ViT backbone in FIBCL-ViT with a ResNet backbone, while keeping all other network components, training strategies, and hyperparameter settings unchanged. This design allows for a fair comparison and isolates the effect of the backbone architecture on overall model performance. The experimental results under three different training-to-testing split ratios (3:7, 5:5, and 7:3) are reported in Table V and Fig. 5.

Under the 3:7 split, the original FIBCL-ViT achieved an accuracy of 0.9167, a precision of 0.9175, a recall of 0.9164, and an AUC of 0.9883, whereas the ResNet-based variant obtained lower performance with an accuracy of 0.9021, a precision of 0.9023, a recall of 0.9005, and an AUC of 0.9799.

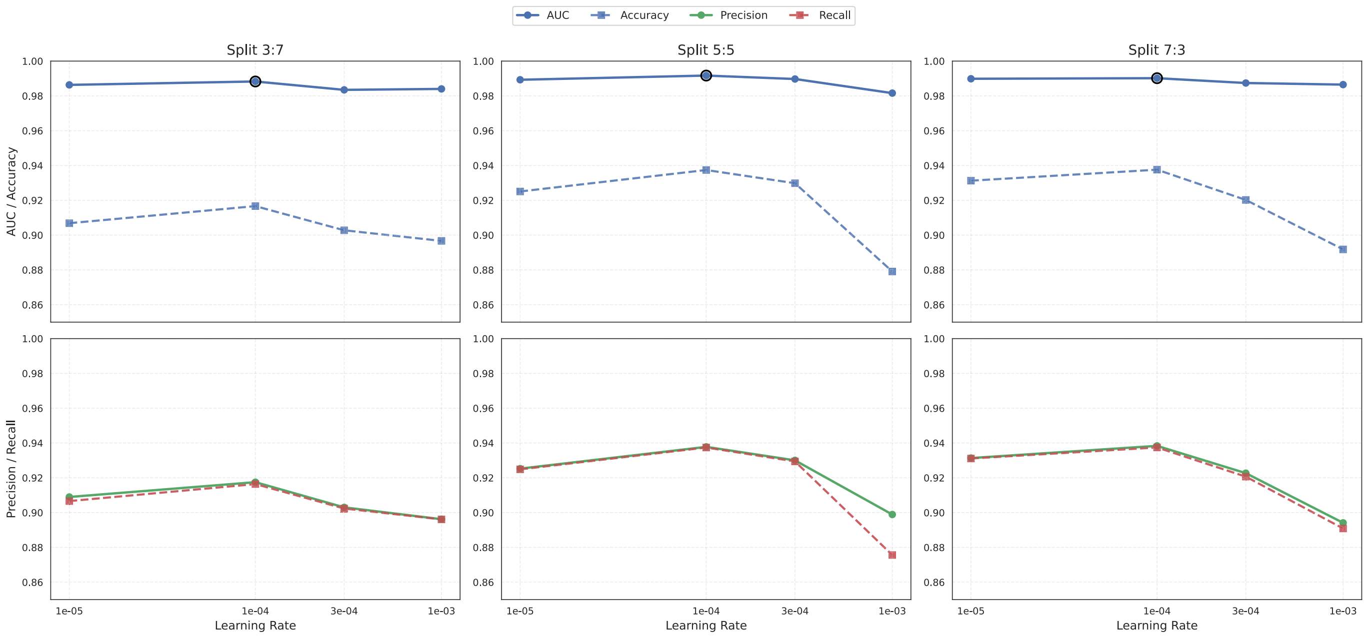

Fig. 6. Performance comparison under different learning rates.

When the split ratio was set to 5:5, FIBCL-ViT further improved its performance to an accuracy of 0.9374 and an AUC of 0.9917, outperforming the ResNet-based model, which achieved an accuracy of 0.9194 and an AUC of 0.9860.

Similarly, under the 7:3 split, FIBCL-ViT maintained its superiority with an accuracy of 0.9376 and an AUC of 0.9902, compared to the ResNet-based variant, which achieved 0.9163 accuracy and 0.9873 AUC.

Overall, across all three data partition settings, FIBCL-ViT consistently outperformed the ResNet-based variant, demonstrating that the transformer-based backbone is more effective in capturing global contextual information and fine-grained discriminative features. These results indicate that incorporating a ViT backbone significantly enhances the representation capability of the FIBCL framework, particularly for medical image classification tasks involving subtle structural variations.

E. Sensitivity Analysis

As shown in Fig. 6 and Table VI, the FIBCL-ViT model was evaluated with learning rates of 1×10^{-5}, 3×10^{-5}, 1×10^{-4}, and 1×10^{-3} under three different data split settings. Across all splits, the model achieved the best and most stable performance when the learning rate was set to 1×10^{-4}, while smaller learning rates led to suboptimal convergence and a larger learning rate caused noticeable performance degradation. This indicates that an appropriate learning rate is crucial for stable optimization and optimal performance.

F. Visualization Analysis

Fig. 7 shows the t-SNE visualizations and corresponding confusion matrices of the FIBCL-ViT model at the 1st, 5th, and 10th training epochs. At early training stages, feature embeddings from different classes exhibit noticeable overlap and higher misclassification rates. As training progresses, interclass separation becomes increasingly clear, and confusion matrices show stronger diagonal dominance. These results indicate that FIBCL-ViT progressively learns more discriminative and class-consistent representations during training.

TABLE VI
Ablation Study on Learning Rate

Learning Rate	Run	AUC	Prec	Rec	Acc
1×10^{-5}	3:7	0.9863	0.9090	0.9066	0.9068
	5:5	0.9893	0.9253	0.9249	0.9251
	7:3	0.9898	0.9313	0.9311	0.9313
	Average	0.9885	0.9219	0.9209	0.9211
3×10^{-5}	3:7	0.9835	0.9030	0.9024	0.9028
	5:5	0.9897	0.9300	0.9294	0.9298
	7:3	0.9874	0.9227	0.9207	0.9202
	Average	0.9869	0.9186	0.9175	0.9176
1×10^{-4}	3:7	**0.9883**	**0.9175**	**0.9164**	**0.9167**
	5:5	**0.9917**	**0.9378**	**0.9374**	**0.9374**
	7:3	**0.9902**	**0.9384**	**0.9374**	**0.9376**
	Average	**0.9901**	**0.9312**	**0.9304**	**0.9306**
1×10^{-3}	3:7	0.9840	0.8962	0.8961	0.8967
	5:5	0.9816	0.8989	0.8756	0.8791
	7:3	0.9865	0.8941	0.8909	0.8918
	Average	0.9840	0.8964	0.8875	0.8892

G. Discussion

Through comparative experiments under multiple data partition settings, ablation studies, and visualization analyses, we systematically evaluated the performance of the FIBCL-ViT model on the fundus image classification task. The results demonstrate that FIBCL-ViT consistently outperforms existing mainstream methods across all evaluation metrics, particularly

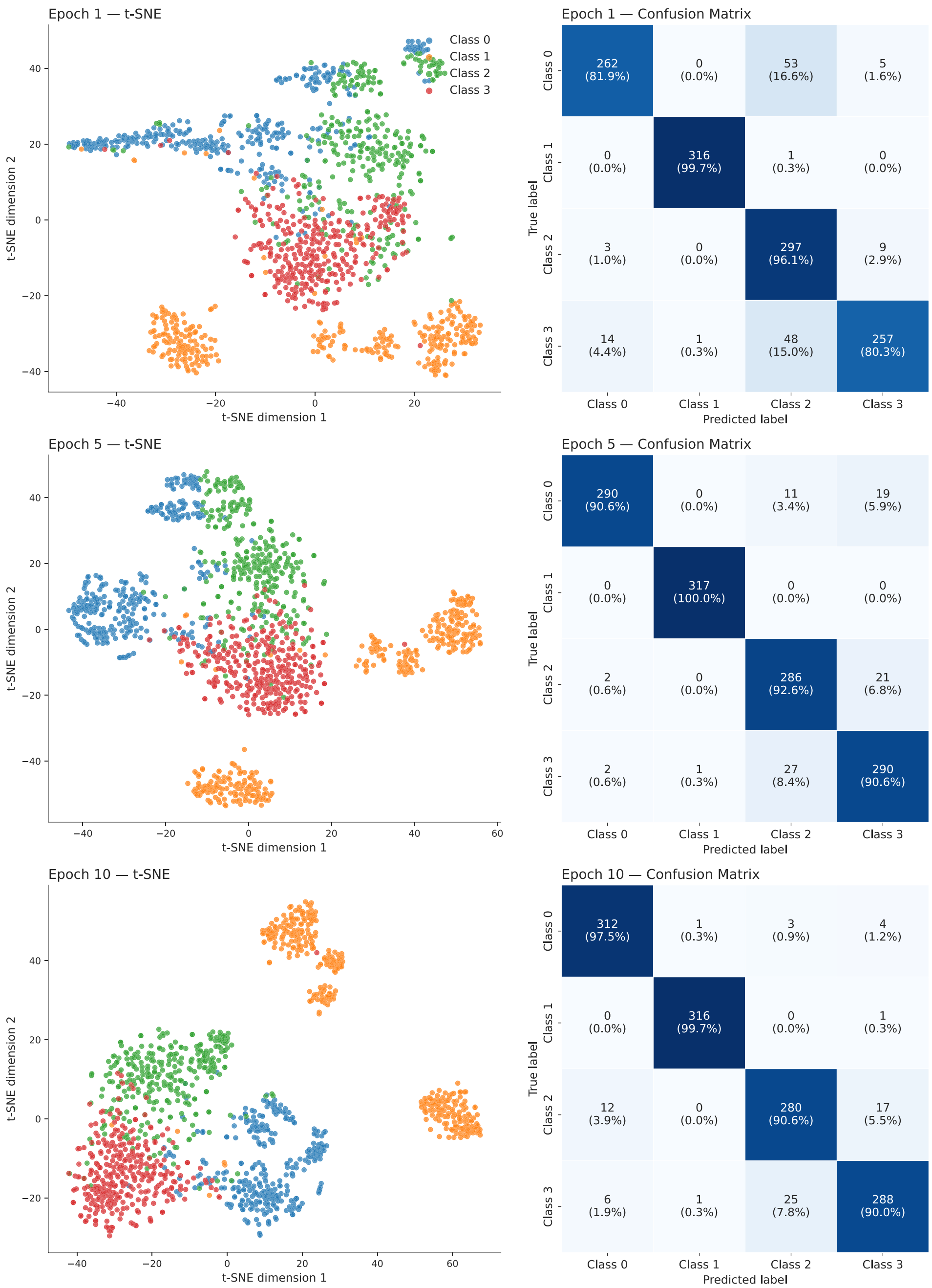

Fig. 7. t-SNE visualization of feature embeddings and confusion matrix at different training iterations.

maintaining strong robustness and generalization capability in scenarios with limited training data. The IB mechanism effectively compresses redundant features, preventing the model from overfitting to irrelevant details, while contrastive learning enhances interclass discriminability, contributing to clearer decision boundaries and improved classification accuracy.

V. Conclusion

This study proposes the FIBCL-ViT model for automated classification and intelligent diagnosis of multiple common ocular diseases from fundus images. The proposed framework integrates the global modeling capability of ViTs, the redundancy-reducing compression of the IB mechanism, and the discriminative enhancement of contrastive learning, thereby achieving high diagnostic accuracy while improving generalization and robustness.

Extensive experiments on a public fundus image dataset demonstrate that FIBCL-ViT consistently outperforms mainstream classification models in terms of accuracy, recall, precision, and AUC, particularly maintaining stable performance under limited training data. Ablation studies and visualization analyses further validate the rationality of the architectural design and show that bottleneck-guided feature compression and contrastive optimization contribute to improved class separability and diagnostic stability.

From the perspective of IoMT applications, the IB mechanism plays a key role in enhancing diagnostic reliability by encouraging compact and task-relevant feature representations while suppressing redundant and noise-sensitive information. This property is particularly beneficial in IoMT environments, where variations in image quality, acquisition conditions, and communication constraints can otherwise degrade model performance.

Despite these encouraging results, several limitations remain. The current study relies on high-quality labeled fundus images and is evaluated on a single curated dataset, which may limit scalability and generalization across different acquisition devices and clinical environments. In addition, computational efficiency and deployment-level performance on resource-constrained IoMT devices have not yet been systematically evaluated.

Future work will focus on validating the proposed model in real-world IoMT scenarios, improving scalability and computational efficiency, reducing dependence on extensive labeled data, and incorporating additional modalities such as OCT and clinical metadata. Expanding the framework to broader disease categories and larger multicenter datasets will further enhance its practical value. In clinical practice, the proposed method has the potential to serve as an auxiliary tool for automated retinal disease screening and remote diagnosis, contributing to improved efficiency and accessibility of ophthalmic care.

REFERENCES

[1] J. M. Forbes and M. E. Cooper, "Mechanisms of diabetic complications," *Physiol. Rev.*, vol. 93, no. 1, pp. 137–188, 2013.

[2] D. S. Fong et al., "Diabetic retinopathy," *Diabetes care*, vol. 26, pp. s99–s102, Oct. 2003.

[3] R. N. Weinreb, T. Aung, and F. A. Medeiros, "The pathophysiology and treatment of glaucoma: A review," *Jama*, vol. 311, no. 18, pp. 1901–1911, 2014.

[4] A. K. Schuster, C. Erb, E. M. Hoffmann, T. Dietlein, and N. Pfeiffer, "The diagnosis and treatment of glaucoma," *Deutsches Ärzteblatt Int.*, vol. 117, no. 13, p. 225, 2020.

[5] H. Jayaram, M. Kolko, D. S. Friedman, and G. Gazzard, "Glaucoma: Now and beyond," *Lancet*, vol. 402, no. 10414, pp. 1788–1801, Nov. 2023.

[6] L. S. Lim, P. Mitchell, J. M. Seddon, F. G. Holz, and T. Y. Wong, "Age-related macular degeneration," *Lancet*, vol. 379, no. 9827, pp. 1728–1738, 2012.

[7] J. Ambati and B. J. Fowler, "Mechanisms of age-related macular degeneration," *Neuron*, vol. 75, no. 1, pp. 26–39, 2012.

[8] D. Pavan-Langston, "Manual of ocular diagnosis and therapy," 1996. [Online]. Available: https://api.semanticscholar.org/CorpusID:56582935

[9] P. P. Shinde and S. Shah, "A review of machine learning and deep learning applications," in *Proc. 4th Int. Conf. Comput. Commun. Control Autom. (ICCUBEA)*, Aug. 2018, pp. 1–6.

[10] Y. Guo, Y. Liu, A. Oerlemans, S. Lao, S. Wu, and M. S. Lew, "Deep learning for visual understanding: A review," *Neurocomputing*, vol. 187, pp. 27–48, Apr. 2016.

[11] A. Voulodimos, N. Doulamis, A. Doulamis, and E. Protopapadakis, "Deep learning for computer vision: A brief review," *Comput. Intell. Neurosci.*, vol. 2018, no. 1, 2018, Art. no. 7068349.

[12] D. A. Forsyth and J. Ponce, *Computer Vision: A Modern Approach*. Upper Saddle River, NJ, USA: Prentice-Hall, 2002.

[13] D. S. W. Ting et al., "Artificial intelligence and deep learning in ophthalmology," *Brit. J. Ophthalmol.*, vol. 103, no. 2, pp. 167–175, 2019.

[14] K. Zhang et al., "An interpretable and expandable deep learning diagnostic system for multiple ocular diseases: Qualitative study," *J. Med. Internet Res.*, vol. 20, no. 11, Nov. 2018, Art. no. e11144.

[15] N. Li, T. Li, C. Hu, K. Wang, and H. Kang, "A benchmark of ocular disease intelligent recognition: One shot for multi-disease detection," in *Proc. Int. Symp. Benchmarking, Measuring Optim.* Springer, 2020, pp. 177–193.

[16] A. Gatouillat, Y. Badr, B. Massot, and E. Sejdic, "Internet of Medical Things: A review of recent contributions dealing with cyber-physical systems in medicine," *IEEE Internet Things J.*, vol. 5, no. 5, pp. 3810–3822, Oct. 2018.

[17] K. Han et al., "A survey on vision transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 87–110, Jan. 2023.

[18] S. Khan, M. Naseer, M. Hayat, S. W. Zamir, F. S. Khan, and M. Shah, "Transformers in vision: A survey," *ACM Comput. Surv.*, vol. 54, no. 10s, pp. 1–41, 2022.

[19] N. Tishby and N. Zaslavsky, "Deep learning and the information bottleneck principle," in *Proc. IEEE Inf. Theory Workshop (ITW)*, Apr. 2015, pp. 1–5.

[20] A. M. Saxe et al., "On the information bottleneck theory of deep learning," *J. Stat. Mech., Theory Exp.*, vol. 2019, no. 12, 2019, Art. no. 124020.

[21] X. Wang and G.-J. Qi, "Contrastive learning with stronger augmentations," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 5, pp. 5549–5560, May 2023.

[22] X. Yang, G. Yang, and J. Chu, "GraphCL-DTA: A graph contrastive learning with molecular semantics for drug-target binding affinity prediction," *IEEE J. Biomed. Health Informat.*, vol. 28, no. 8, pp. 4544–4552, Aug. 2024.

[23] M. J. Umer, M. Sharif, M. Raza, and S. Kadry, "A deep feature fusion and selection-based retinal eye disease detection from oct images," *Expert Syst.*, vol. 40, no. 6, p. 13232, Jul. 2023.

[24] A. Albelaihi and D. M. Ibrahim, "DeepDiabetic: An identification system of diabetic eye diseases using deep neural networks," *IEEE Access*, vol. 12, pp. 10769–10789, 2024.

[25] A. Adel, M. M. Soliman, N. E. M. Khalifa, and K. Mostafa, "Automatic classification of retinal eye diseases from optical coherence tomography using transfer learning," in *Proc. 16th Int. Comput. Eng. Conf. (ICENCO)*, Dec. 2020, pp. 37–42.

[26] P. Glaret subin and P. Muthukannan, "Optimized convolution neural network based multiple eye disease detection," *Comput. Biol. Med.*, vol. 146, Jul. 2022, Art. no. 105648.

[27] O. Bernabé, E. Acevedo, A. Acevedo, R. Carreño, and S. Gómez, "Classification of eye diseases in fundus images," *IEEE Access*, vol. 9, pp. 101267–101276, 2021.

[28] D. S. W. Ting et al., "Development and validation of a deep learning system for diabetic retinopathy and related eye diseases using retinal images from multiethnic populations with diabetes," *JAMA*, vol. 318, no. 22, pp. 2211–2223, Dec. 2017.

[29] A. Elsawy et al., "Multidisease deep learning neural network for the diagnosis of corneal diseases," *Amer. J. Ophthalmol.*, vol. 226, pp. 252–261, Jun. 2021.